PULLBACK ALONG THE PROJECTION OF THE VECTOR BUNDLE OF AN INVERTIBLE SHEAF IS AN ISOMORPHISM ON CHOW GROUPS, BY WAY OF THE ZERO SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Surjectivity of the flat pullback along a bundle whose fibres are affine spaces. Same chapter chow.tex, label lemma-pullback-affine-fibres-surjective (invoked at proof line 5753). It gives the surjectivity half of the isomorphism.
- The cap with the class of a relative effective Cartier divisor as a left inverse of the pushforward along the projection. Same chapter chow.tex, label lemma-relative-effective-cartier (invoked at proof line 5758). Since $o(X)$ is an effective Cartier divisor on L, it gives $o^* \circ p^* = \text{id}$, which forces injectivity of p^*.

Source reference. Original lemma 02TT: sources/stacks-chow/pinned-original-source.tex, source lines 5651–5651. Navigation pointer for \ref{lemma-pullback-affine-fibres-surjective} used inside the counted span (line 5753); the argument that consumes it is retained in the span.

Source reference. Original lemma 02TR: sources/stacks-chow/pinned-original-source.tex, source lines 5598–5598. Navigation pointer for \ref{lemma-relative-effective-cartier} used inside the counted span (line 5758); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup}

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used inside the counted span (line 5738); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let*

$$p : L = \underline{\mathrm{Spec}}(Sym^*(\mathcal{L})) \longrightarrow X$$

be the associated vector bundle over X. Then $p^* : \mathrm{CH}_k(X) \to \mathrm{CH}_{k+1}(L)$ is an isomorphism for all k.

Proof. For surjectivity see Lemma 02TT. Let $o : X \to L$ be the zero section of $L \to X$, i.e., the morphism corresponding to the surjection $\mathrm{Sym}^*(\mathcal{L}) \to \mathcal{O}_X$ which maps $\mathcal{L}^{\otimes n}$ to zero for all $n > 0$. Then $p \circ o = \mathrm{id}_X$ and $o(X)$ is an effective Cartier divisor on L. Hence by Lemma 02TR we see that $o^* \circ p^* = \mathrm{id}$ and we conclude that p^* is injective too. □